Mock exam 3

Planetary Systems final exam practice • closed book • 120 minutes

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This mock exam has the same shape and level as the final exam: 6 questions of 10 points each, 60 points in total, in a 120-minute closed-book sitting where a calculator and a pen are all that is allowed. Your grade is $1+0.15\times$points, rounded to the nearest half grade; the pass mark is 30 points. The twelve relations on the exam formula list are assumed known; every other formula a question needs is printed in the question itself. Marks per part are shown in brackets, and the steps carry marks, not only the number: show your algebra. Some parts state an intermediate result ("show that …"); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps and round only at the end.

Constants and data

$G = 6.674\times10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ 1 AU = 1.496×10^{11} m 1 yr = 3.156×10^{7} s = 365.25 d
1 d = 86 400 s $k_B = 1.381\times10^{-23}\ \mathrm{J\,K^{-1}}$ $m_u = 1.661\times10^{-27}$ kg $\sigma = 5.670\times10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$
$S_0 = 1361\ \mathrm{W\,m^{-2}}$ (solar constant at 1 AU)

Body	a, A	Further data
Sun		$R = 6.957\times10^{5}$ km
Jupiter		$R = 69\,911$ km
Callisto		orbits Jupiter with $a = 1.883\times10^{6}$ km, P = 16.69 d
Earth	1.000 AU, $A = 0.30$	P = 1 yr, $g = 9.81\ \mathrm{m\,s^{-2}}$, surface T = 288 K, mean molecular weight $\mu = 28.97$
Moon		$g = 1.62\ \mathrm{m\,s^{-2}}$
Mercury		$R = 2440$ km, $\bar{\rho} = 5427\ \mathrm{kg\,m^{-3}}$
Titan		surface T = 94 K, surface pressure 1.5×10^{5} Pa, methane mole fraction 5%

A total power L spread over a sphere of radius r gives a flux $L/(4\pi r^2)$; for sunlight this means $S = S_0/a^2$ with a in AU. A body of mass m moving at speed v carries the kinetic energy $E = \frac{1}{2}mv^2$. Dry air on Earth cools by 9.8 K per kilometre of ascent (the dry adiabatic lapse rate).

Problem 1 Weighing the Sun and Jupiter

An orbit is a balance you can read a mass from: the central mass alone sets how fast a small companion circles at a given distance. This problem uses only relations from the formula list.

(a) **[4 pts]** Use the Earth's orbit (data table) to compute the mass of the Sun in kilograms.

(b) **[3 pts]** Use Callisto's orbit (data table) to compute the mass of Jupiter in kilograms.

(c) **[3 pts]** Compute the mean density of the Sun and of Jupiter.

Problem 2 Reading a cratered surface

Impact craters are both a hazard record and a clock. In the gravity regime, an impact of energy E into a target of density ρ_t and surface gravity g opens a crater of diameter

$$D \approx \left(\frac{E}{\rho_t\, g}\right)^{1/4}$$

(all quantities in SI units, D in metres). Craters larger than a transition diameter D_t collapse into complex craters with central peaks and terraces; on the Moon $D_\mathrm{t} \approx 15$ km. For surfaces younger than about 3 Gyr, the cumulative density of craters larger than 1 km grows linearly with age T:

$$N(\geq 1\ \mathrm{km}) \approx b\,T, \qquad b = 8.38 \times 10^{-4}\ \mathrm{km}^{-2}\,\mathrm{Gyr}^{-1}.$$

(a) **[3 pts]** A rocky asteroid of diameter 1.2 km and density 3000 kg m^{-3} hits at 18 km s^{-1}. Show that its kinetic energy is 4.397×10^{20} J.

(b) **[3 pts]** This asteroid strikes the Moon, where the target rock has density 2500 kg m^{-3}. Compute the crater diameter and state whether the crater is simple or complex.

(c) **[4 pts]** A lava plain on the Moon shows $N(\geq 1\ \mathrm{km}) = 1.8 \times 10^{-3}$ km^{-2}. Compute its age, check that the linear relation applies, and explain in 1 to 2 sentences why crater counting cannot date the oldest lunar highlands this way.

Problem 3 The birth of a core and a dynamo

Terrestrial planets separate into an iron core and a rocky mantle early in their history, and a convecting liquid core can drive a magnetic dynamo. A convecting core of flow speed U, size L, and magnetic diffusivity η sustains a dynamo when the magnetic Reynolds number

$$\mathrm{Rm} = \frac{UL}{\eta}$$

exceeds a critical value of order 10 to 100.

(a) **[4 pts]** The radioactive isotope ^{182}Hf decays to ^{182}W with a half-life of 8.9 Myr and is the main clock for dating core formation. Compute the decay constant λ, the fraction of the initial ^{182}Hf remaining 30 Myr after the birth of the solar system, and explain in 1 to 2 sentences why this clock cannot date events later than about 45 Myr.

(b) **[3 pts]** Early Mars had a convecting liquid core with $U = 5 \times 10^{-4}$ m s^{-1}, $L = 1.8 \times 10^{6}$ m, and $\eta = 1$ m^2 s^{-1}. Compute the magnetic Reynolds number and state whether a dynamo is expected then, and what happens once core convection stops.

(c) **[3 pts]** Judge this claim in 2 to 3 sentences: *"Mars has no global magnetic field today, so it can never have had one."*

Problem 4 Cloud bases on Earth and Titan

Clouds form where a gas becomes saturated. The saturation vapour pressure over a condensed phase follows the integrated Clausius-Clapeyron relation

$$P_{\mathrm{sat}}(T) = P_{\mathrm{ref}} \exp\left[-\frac{L_v}{R_v}\left(\frac{1}{T} - \frac{1}{T_{\mathrm{ref}}}\right)\right],$$

where L_v/R_v is a constant of the condensing species. For water, $L_v/R_v = 5400$ K with reference point $(T_{\mathrm{ref}}, P_{\mathrm{ref}}) = (273\ \mathrm{K},\ 611\ \mathrm{Pa})$. For methane, $L_v/R_v = 980$ K with reference point $(90.7\ \mathrm{K},\ 1.17 \times 10^4\ \mathrm{Pa})$.

(a) **[3 pts]** Compute the pressure scale height of the Earth's atmosphere near the surface (data table).

(b) **[3 pts]** An air parcel at the surface has temperature 303 K and water vapour pressure 1.7 kPa. Compute the dew point of the parcel, and use the dry adiabatic lapse rate to estimate the altitude of the cloud base it forms when it rises.

(c) **[4 pts]** On Titan, methane plays the role of water. Compute the saturation vapour pressure of methane at Titan's surface temperature, compare it with the actual methane partial pressure at the surface (data table), and explain in 1 to 2 sentences what this means for Titan's lakes and clouds.

Problem 5 Mercury, the shrinking planet

Mercury is scarred by long cliffs (lobate scarps) that record a global contraction of the planet as its interior cooled. Cooling the whole planet by ΔT releases the heat $E = Mc\,\Delta T$, with specific heat capacity $c = 800\ \mathrm{J\,kg^{-1}\,K^{-1}}$, and shrinks the radius by

$$\Delta R = \frac{\alpha_V}{3} R\,\Delta T, \qquad \alpha_V = 3 \times 10^{-5}\ \mathrm{K^{-1}}.$$

(a) **[4 pts]** Compute Mercury's mass from its radius and mean density, and show that cooling the whole planet by $\Delta T = 200$ K releases 5.283×10^{28} J.

(b) **[3 pts]** Assume this heat left the planet at a constant rate over 4.5 Gyr. Compute the mean surface heat flow in $\mathrm{mW\,m^{-2}}$ and compare it with the Earth's present mean surface heat flow of 87 $\mathrm{mW\,m^{-2}}$.

(c) **[3 pts]** Compute the radius change from the 200 K cooling and compare it with the 7 km of contraction inferred from Mercury's lobate scarps.

Problem 6 A planet around a K dwarf

Stars less massive than the Sun burn slower and live longer. On the main sequence, luminosity scales with stellar mass as

$$\frac{L}{L_\odot} = \left(\frac{M}{M_\odot}\right)^{3.5},$$

and Kepler's third law for an orbit around a star of mass M reads, in convenient units,

$$P^2 = \frac{a^3}{M/M_\odot} \qquad (P \text{ in yr},\ a \text{ in AU}).$$

Consider a K dwarf of mass $M = 0.7\ M_\odot$.

(a) **[3 pts]** Compute the luminosity of the K dwarf in solar units.

(b) **[4 pts]** Show that a planet receives the same flux as the Earth at $d = 0.5357$ AU from this star, and verify that with the Earth's albedo it has the Earth's equilibrium temperature.

(c) **[3 pts]** Compute the orbital period of this planet. Then judge this claim in 2 to 3 sentences: *"The smaller the star, the better for life, because the star lives longer."*